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Master Thesis

Influence of atmospheric vertical wind and turbulence on noctilucent clouds

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Abstract

Noctilucent clouds (NLCs) form near the cold summer mesopause and are extremely sensitive to changes in temperature, water vapour and atmospheric dynamics. In this work, I investigate how two key dynamical processes, small scale turbulence and large scale vertical wind control NLC microphysics and the distribution of water vapour above $69°N$ during July. The study uses the LIMA general circulation model together with the Lagrangian microphysical model MIMAS.

The results show that increased eddy diffusivity mixes the atmosphere more strongly and leads to NLCs that are diffuse. In fact, turbulence act as a bouncer at the lower boundary of NLC growth region, by kicking ice particles out of growth zone. This reduces their growth, lowers particle number density, and noticeably weakens NLC brightness. Vertical wind behaves very differently: stronger upwelling brings more water vapour into the cold ice-forming region and slows down sedimentation. This produces larger particles, brighter clouds, and clear changes in the vertical structure of the layer.

Both effects are consistent with previous observations from LIDAR and satellite climatologies, as well as laboratory studies of ice growth under mesospheric conditions. The findings highlight that turbulence and vertical wind influence NLCs through different physical pathways, and that an accurate representation of these transport processes is essential for understanding NLC variability and their connection to the changing atmosphere.

Zusammenfassung

Noctilucente Wolken (NLCs) bilden sich nahe der kalten Sommermesopause und sind äußerst empfindlich gegenüber Änderungen von Temperatur, Wasserdampf und atmosphärischer Dynamik. In dieser Arbeit untersuche ich, wie zwei wichtige dynamische Prozesse – kleinräumige Turbulenz und großräumiger Vertikalwind – die Mikrophysik von NLCs und die Verteilung von Wasserdampf über $69°N$ im Juli steuern. Die Studie verwendet das LIMA-Allgemeine Zirkulationsmodell zusammen mit dem Lagrange'schen mikrophysikalischen Modell MIMAS.

Die Ergebnisse zeigen, dass erhöhte Eddy-Diffusivität die Atmosphäre stärker durchmischt und zu diffuseren NLCs führt. Tatsächlich wirkt Turbulenz wie ein Türsteher an der unteren Grenze des NLC-Wachstumsbereichs, indem sie Eisteilchen aus der Wachstumszone herausstößt. Dies reduziert ihr Wachstum, verringert die Teilchendichte und schwächt die Helligkeit der NLCs deutlich. Der Vertikalwind verhält sich ganz anders: Stärkerer Auftrieb bringt mehr Wasserdampf in die kalte Eisbildungsregion und verlangsamt die Sedimentation. Dadurch entstehen größere Partikel, hellere Wolken und klare Veränderungen in der vertikalen Struktur der Schicht.

Beide Effekte stimmen mit früheren Beobachtungen aus LIDAR- und Satelliten-Klimatologien sowie mit Laborstudien zum Eiswachstum unter mesospherischen Bedingungen überein. Die Ergebnisse zeigen, dass Turbulenz und Vertikalwind NLCs über unterschiedliche physikalische Mechanismen beeinflussen und dass eine genaue Darstellung dieser Transportprozesse entscheidend ist, um die Variabilität von NLCs und ihre Verbindung zur sich verändernden Atmosphäre zu verstehen.

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Acknowledgement

This thesis represents far more than an academic requirement; it is a reflection of the journey I lived through in Hamburg, the people who stood beside me, and the moments that shaped me into who I am today.

When I first arrived in Germany, everything felt unfamiliar and overwhelming. Yet, it was during these uncertain days that the Leibniz Institute of Atmospheric Physics (IAP) became the place that grounded me. It was not merely an institute; it became the space where my ideas turned into purpose, where my interests found direction, and where I discovered what kind of researcher I wanted to become.

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This achievement is not mine alone; it belongs to everyone who walked with me, supported me, encouraged me, and believed in me. I carry your contributions with gratitude and pride.

Contents

Nomenclature

Abbreviations

Abbreviation	Definition
ALOMAR	Arctic Lidar Observatory for Middle Atmosphere Research (69°N)
AIM	Aeronomy of Ice in the Mesosphere (NASA satellite mission)
CIPS	Cloud Imaging and Particle Size experiment (AIM UV imager)
CN	Condensation nuclei
H_2O	Water vapour
CH_4	Methane (source of mesospheric H_2O through oxidation)
CO_2	Carbon dioxide (long-term cooling agent in the mesosphere)
GW	Gravity waves
IWC	Ice water content
LIMA	Leibniz Institute Middle Atmosphere general circulation model
Ly-α	Solar Lyman-α emission at 121.6 nm
MIMAS	Mesospheric Ice Microphysics And Transport model
MLT	Mesosphere and Lower Thermosphere
MSP	Meteoric smoke particle(s)
NLC	Noctilucent clouds (ground-based term)
PMC	Polar mesospheric clouds (satellite term for NLCs)
PMSE	Polar Mesospheric Summer Echoes (VHF/UHF radar echoes)
SBUV	Solar Backscatter Ultraviolet instrument (long-term PMC record)
SOFIE	Solar Occultation For Ice Experiment (AIM IR instrument)

Symbols

Symbol	Definition
β_{532}	Volume backscatter coefficient at 532 nm (proxy for NLC brightness)

Symbol	Definition
K_{zz}	Eddy diffusion coefficient for vertical turbulent mixing ($\mathrm{m^2\,s^{-1}}$)
n_{ice}	Ice-particle number density ($\mathrm{cm^{-3}}$)
r_{ice}	Mean ice-particle radius (nm)
S	Water-vapour supersaturation ratio ($P_{\mathrm{H_2O}}/P_{\mathrm{ice}}$)
T	Temperature (K)
θ	Potential temperature (K)
u, v	Zonal and meridional wind components ($\mathrm{m\,s^{-1}}$)
w	Vertical wind velocity from LIMA ($\mathrm{m\,s^{-1}}$)
w_{bi}	Background vertical velocity acting on particles ($\mathrm{m\,s^{-1}}$)
w_{ei}	Eddy-induced vertical velocity ($\mathrm{m\,s^{-1}}$)
w_{si}	Sedimentation velocity of ice particles ($\mathrm{m\,s^{-1}}$)
z	Altitude (km)
φ	Latitude (degrees)
ρ	Air density ($\mathrm{kg\,m^{-3}}$)

1 Introduction

Noctilucent clouds (NLCs) are among the most striking and mysterious phenomena the atmosphere can display. On clear summer nights at high latitudes, thin, silver blue clouds suddenly appear low on the poleward horizon, long after the tropospheric clouds have faded into darkness. The Sun is already well below the horizon, the lower atmosphere is in twilight or fully dark, and yet this fragile layer is still glowing because it sits so high that it remains sunlit from below. On good nights, the patterns ripple, twist, and break into bands that look almost like a frozen sea overhead.

NLCs were first identified in the late nineteenth century. The first reliable reports date back to 1885, shortly after the Krakatoa eruption, when observers in northern Europe began systematic night-sky watching and noticed unfamiliar, luminous clouds that did not fit normal weather classifications (Jesse, 1885). For a long time they were treated mainly as a curiosity: logged in observing catalogues, photographed, and described, but not yet fully understood. By the mid–twentieth century, careful analyses of their apparent heights, unique height, and seasonal occurrence gradually shifted the focus from description to physics (Witt, 1962). The basic questions that motivated those early studies were: What are NLCs made of? Where and when do they form? And which processes control their occurrence and brightness?

Today we know that noctilucent clouds are the highest clouds in Earth's atmosphere and form a very thin, seasonal layer near the summer mesopause. Because they sit at the very edge of the water–ice phase space, they are not only visually impressive but also extremely sensitive to changes in temperature, water vapour, and dynamics. That makes them both a beautiful phenomenon and a potential indicator of long–term changes in the middle atmosphere.

NLCs form near the cold summer mesopause, typically between about 82 and 86 km altitude (Kiliani et al., 2013). In satellite records they are referred to as polar mesospheric clouds (PMCs), but physically they are the same phenomenon. The clouds consist of tiny water–ice particles with radii of only a few tens of nanometres (Cossart et al., 1999; Rapp & Thomas, 2006). They become visible because they efficiently forward scatter sunlight at very shallow solar elevations, while the air below is already in shadow. During daytime, the clouds may still exist, but Rayleigh scattering in the lower atmosphere hides them from view (Gadsden, 1990).

Ice formation in this region is not a straightforward process. Temperatures must be extremely low, water vapour must be present in sufficient amounts, and suitable condensation nuclei must exist. Early works already recognised that NLCs are tenuous ice clouds that form under strong supersaturation of water near the cold summer mesopause (Turco et al., 1982.

Figure 1.1: Noctilucent clouds spotted from Kühlungsborn, captured by Dr. Ashique Vellalassery in summer 2025. The lower atmosphere is already dark, but the cloud layer near the mesopause is still illuminated by the Sun from below the horizon.

Studies showed that the dominant condensation nuclei are meteoric smoke particles (MSPs), nanometre scale debris created when meteoroids ablate and recondense between roughly 70 and 100 km (Hunten et al., 1980). Global modelling placed MSPs throughout the upper mesosphere with a clear seasonal and latitudinal structure (Megner et al., 2006; Megner et al., 2008), and limb occultation measurements from space finally detected MSP material inside ice particles in PMCs (Hervig et al., 2012). This strongly supports heterogeneous nucleation on meteoric material as the main pathway, while homogeneous nucleation is thought to be rare because it demands unrealistically high supersaturation (Rapp & Thomas, 2006; Wilms et al., 2016).

Once a dust particle reaches the suitable condition, it grows by vapour deposition whenever the local saturation ratio $S = P_{\mathrm{H_2O}}/P_{\mathrm{sat,ice}}(T)$ (where P_{H_2O} and P_{ice} partial pressure of water vapour and saturation water vapour over a flat ice surface) exceeds one, shrinks by sublimation when $S < 1$, and slowly settles downward under gravity, all the while being advected and mixed by the surrounding flow. Figure 1.2 summarises this life cycle: nucleation on MSPs, growth by vapour uptake, sedimentation, turbulent transport, and eventual evaporation (Rapp & Thomas, 2006; Kiliani et al., 2013). The relative strength and timing of these processes determine when and where NLC layers appear, how bright they

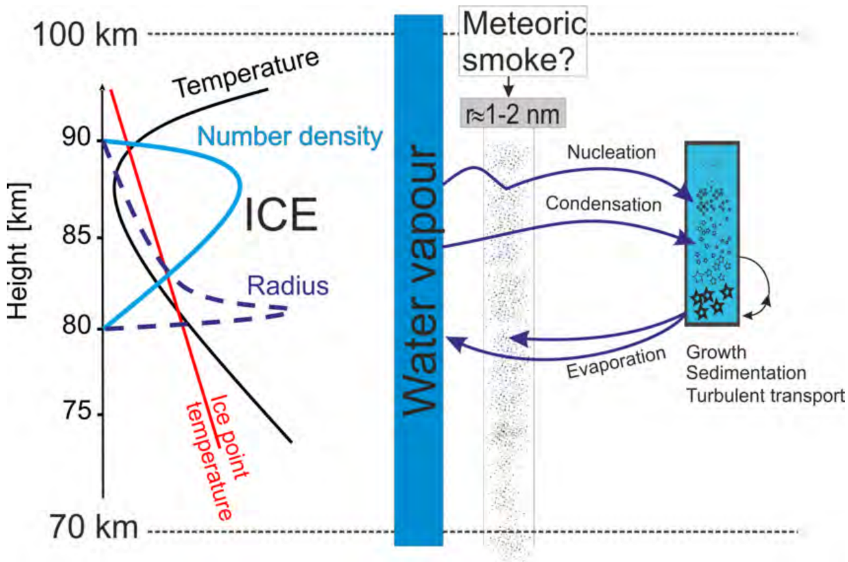


Figure 1.2: Summary of the main microphysical processes that control noctilucent clouds: nucleation on meteoric smoke particles, growth by vapour deposition, sedimentation, turbulent mixing, and sublimation (adapted from Kiliani et al., 2013).

become, and how they respond to changes in the background atmosphere.

To understand why NLCs form only in a very narrow altitude and season, it is useful to look at the thermal structure of the atmosphere as a whole. The atmosphere is a thin layer of gases and suspended particles surrounding Earth. Its density decreases roughly exponentially with height, so that pressure drops by about one order of magnitude every ~ 16 km. Long before we reach the conventional "edge of space" near 100 km, the air has already become extremely tenuous.

Solar short–wave radiation, absorbed mainly by ozone and molecular oxygen, and long–wave radiation emitted upward from the surface, combine to produce a temperature profile that is far from monotonic. Instead, a series of temperature minima and maxima divide the atmosphere into layers: the troposphere, stratosphere, mesosphere, and thermosphere (see 1.3). Each layer has its own characteristic heating and cooling mechanisms and therefore its own circulation.

The lowest layer, the troposphere, extends from the surface to about 10–18 km, contains most of the atmospheric mass and almost all of its water vapour, and is where all familiar weather events takes place (University Corporation for Atmospheric Research (UCAR), 2015). Temperatures are warmest at the ground and decrease upward at a typical rate of about $7\,\mathrm{K\,km^{-1}}$. Above that, in the stratosphere (up to ~ 50 km), temperatures increase with height because ozone absorbs ultraviolet radiation more efficiently than the layer can radiatively cool.

Beyond the stratopause, temperatures fall again through the mesosphere, which extends to about 90 km. Here, the gas is too thin to absorb much solar energy directly and is instead largely heated from below (Oceanic & (NOAA), 2024). The temperature reaches a global minimum near the mesopause. In polar summers, this minimum can drop to 120 K or even lower (Kiliani et al., 2013; Lübken, 1999), crossing far below the frost point of water vapour

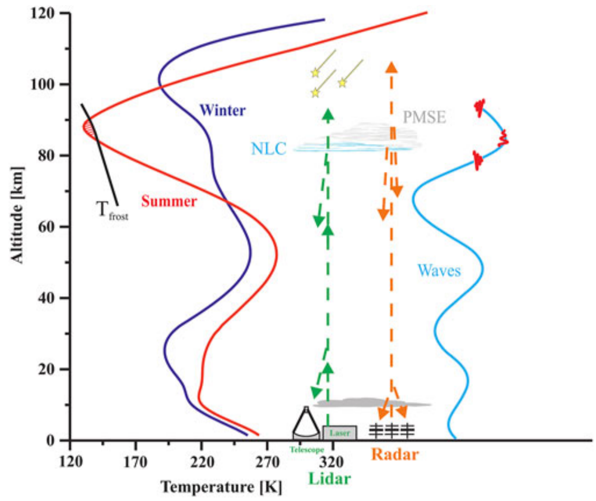


Figure 1.3: Schematic of the thermal structure of the middle atmosphere (adapted from Lübken et al., 2021). Red and blue curves show mean summer and winter temperature profiles; the slanted black line marks the ice–frost point. Their intersection near the cold summer mesopause (~80–90 km, $T \lesssim 150$ K) defines the altitude range where water ice can exist and NLCs can form. The cyan curve sketches the increase of gravity–wave amplitudes with height and breaking near the mesopause.

and making ice formation possible.

The mesosphere and lower thermosphere (MLT) region, roughly 50–100 km, is therefore key for NLCs. It sits between the weather dominated atmosphere below and near–space above, hosts the coldest temperatures on Earth, and is strongly influenced by waves and tides. At the same time, it is notoriously difficult to observe: weather balloons cannot reach it, satellites often struggle with non–LTE radiative transfer, and sounding rockets provide only short snapshots. Continuous measurements rely on a small number of resonance lidars, meteor radars, and fallingsphere flights (Lübken et al., 2016; Baumgarten et al., 2010). This sparse observational coverage is one of the main reasons why we still need modelling to understand NLCs in detail.

The extremely low temperatures of the summer mesopause cannot be explained by radiative processes alone. A major contribution comes from the dissipation of atmospheric gravity waves. These waves are buoyancy-driven oscillations that occur when air parcels in a stably stratified fluid are displaced from equilibrium. Gravity waves are continuously generated in the lower atmosphere by topography, deep convection, jet front systems, and other dynamical perturbations (Fritts & Alexander, 2003). Once excited, they propagate upward, and their amplitudes increase as the background density decreases, approximately scaling with $\rho^{-1/2}$, where ρ being atmospheric density, through conservation of wave action.

As they reach the upper mesosphere, many waves become unstable and dissipate through convective or shear instability. This dissipation deposits momentum into the mean flow, commonly referred to as gravity wave drag which drives the meridional residual circulation. The resulting upwelling in the summer hemisphere produces strong adiabatic cooling, effectively setting the cold temperatures characteristic of the summer mesopause. A schematic representation of gravity wave structure is shown in Figure 1.3.

From the NLC perspective, gravity waves therefore play two roles at once, (i) they help maintain the cold background temperatures and upwelling that make ice formation possible, and (ii) they create local temperature minima and vertical displacements that modulate the saturation ratio S in time and height. Wave breaking also links directly to the two dynamical quantities studied in this thesis: it feeds energy into small scale turbulence (controlling eddy diffusivity, K_{zz}) and contributes to the spectrum of vertical wind w that transport both vapour and ice particles.

Because NLCs form only when the temperature is extremely low and the air is sufficiently moist, they naturally highlight regions where the atmosphere comes close to the water–ice phase boundary. In that sense, NLCs are not just an optical curiosity, but also a tracer of the state of the summer mesopause (Thomas, 2003).

Modern observing systems have pushed this idea much further. Ground based Rayleigh-Mie-Raman lidars, such as those at ALOMAR, provide high resolution vertical profiles of backscatter and allow long term monitoring of NLC height, occurrence, and brightness (Baumgarten et al., 2007). Multi wavelength techniques retrieve information on mean radius and size distribution directly from the lidar signal (Cossart et al., 1999). From space, the AIM (Aeronomy of Ice in the Mesosphere) mission opened a global view: the CIPS (Cloud Imaging and Particle Size) instrument maps NLC occurrence and morphology in the UV, while companion measurements constrain temperature and composition relevant to ice microphysics. Together, these datasets have built up a detailed climatology of NLC altitude, seasonality, and radiative properties (DeLand & Thomas, 2015).

NLCs live in a region where gravity waves, tides, and planetary waves strongly modulate temperature and vertical motion. Satellite imagery and ground records show banded and ripple structures in NLCs (it is visibe in figure 1.1) that can be traced back to underlying wave fields (Baumgarten et al., 2012; Chandran et al., 2010; Chandran et al., 2012; Fiedler et al., 2017). Complementary radar observations of polar mesospheric summer echoes (PMSE) reveal how charged ice particles couple to turbulence and small scale dynamics (Rapp & Lübken, 2004). All this means that NLCs do not only reflect the slow, climatological state of the mesosphere, but also it is a rapidly varying wave environment.

However, several key quantities remain hard to measure directly. Continuous profiles of vertical wind and turbulent mixing in the 80–90 km region are still scarce, and most observational inferences are indirect. Microphysical retrievals rely on assumptions about particle habit and size distribution, while global satellite views trade vertical resolution for coverage. As a result, the separate roles of vertical motion and turbulent mixing in particle size, number density, and optical backscatter are not yet uniquely constrained.

Over the last decades, satellite records have shown significant changes in the occurrence and brightness of polar mesospheric clouds (DeLand & Thomas, 2015), and at the same time, greenhouse gases such as CO_2 and CH_4 have been increasing. Modelling studies suggest that higher CO_2 cools the summer mesopause, while methane oxidation moistens the upper mesosphere; together these changes shift the background against which NLCs form (Lübken et al., 2018; Lübken et al., 2021). This has led to the idea that NLCs might act as a sensitive indicator of anthropogenic influence in the middle atmosphere (Zahn, 2003; Fritts & Alexander, 2003).

There is evidence that the gravity wave environment is also changing, or at least that its effects on the mesosphere are highly variable in time. Gravity waves generated in the troposphere and stratosphere are filtered by changing wind and temperature structures on their way up, and their breaking levels can shift as the climate evolves. Because wave breaking controls both upwelling and small–scale turbulence, any change in wave filtering can feed back on the conditions for NLC formation.

All of this points in the same direction: if we want to use NLCs as tracers of the middle atmosphere and its trends, we need to understand how they respond to changes in both vertical motion and turbulent mixing. Simply looking at brightness trends is not enough; we must also know how sensitive the clouds are to reasonable variations in K_{zz} and w under realistic summer conditions.

1.1. Road map of this thesis

The central idea of this thesis is upfront: noctilucent clouds are governed by microphysical processes, but these processes are fundamentally controlled by the dynamics of the summer mesopause. Two dynamical factors are particularly important. The first is K_{zz}, which regulates how efficiently water vapour and ice particles are vertically mixed.The second is w, which determines the upward or downward transport of air masses over longer time scales. Both quantities are closely tied to gravity wave propagation and breaking; where gravity waves dissipate strongly influences the vertical structure of K_{zz} and the strength of vertical winds (Fritts & Alexander, 2003; Lübken, 1997). Long term changes in the background circulation driven, for example, by increasing greenhouse gases are also expected to modify these two dynamical drivers (Lübken et al., 2018).

In this work, a combined modelling framework is used to investigate how noctilucent clouds respond when turbulence and vertical wind are systematically varied. The goal is to quantify how these dynamical processes regulate ice particle growth, sedimentation, water vapour redistribution, and ultimately the optical brightness of NLCs.

The thesis is organised as follows. Chapter 2 describes the modelling framework. It introduces the LIMA general circulation model and the MIMAS microphysical and Lagrangian transport scheme, and outlines the experimental modifications applied to turbulence and vertical wind. The treatment of water vapour, the particle ensemble configuration, and the numerical implementation of particle transport and growth are also explained.

Chapter 3 presents the results. We first examine how turbulence affects the saturation ratio, water vapour structure, ice number density, particle radius, and cloud brightness, including the altitude dependent sensitivity experiments. We then analyse the role of vertical wind, focusing on its impact on water vapour transport, the vertical placement of the ice formation region, and the resulting changes in NLC microphysical and optical properties.

Finally, Chapter 4 summarises the main findings, discusses the limitations of the modelling approach, and outlines how these results can support future observational and modelling efforts aimed at using NLCs as tracers of mesospheric dynamics and climate variability.

2
Methods

This study couples two established tools to simulate noctilucent clouds and their environment. The large scale state of the Northern Hemisphere mesosphere is provided by the *Leibniz Institute Middle Atmosphere* (LIMA) general circulation model. Background dynamics, including temperature, wind, pressure and density fields representative of summer polar conditions (Lübken et al., 2009; Lübken et al., 2013) are obtained from LIMA. These fields drive the *Mesospheric Ice Microphysics and Transport* (MIMAS) model, a Lagrangian scheme that follows individual meteoric smoke particles as they nucleate ice, grow, sublimate and sediment. The overall workflow ; LIMA for background dynamics, MIMAS for cloud microphysics is sketched in Fig. 2.1. In this chapter we will explain the numerical models, the science behind it and the experiment set-up which we used to investigate the objectives of this study.

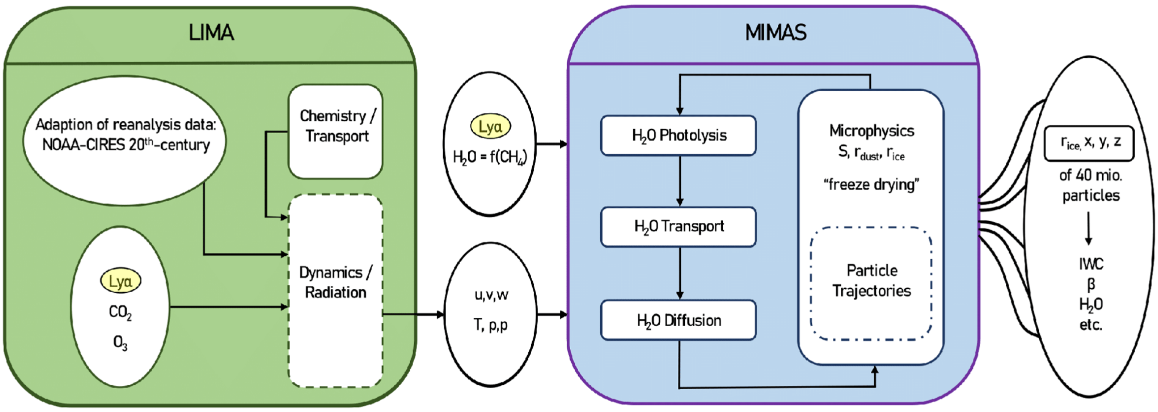


Figure 2.1: Sketch of the LIMA (green) and MIMAS (blue) models (taken from Vellalassery et al., 2023)

2.1. LIMA

LIMA is a non linear, global, three-dimensional Eulerian grid point model that spans the entire neutral atmosphere, from the surface to roughly 150 km altitude (Berger, 2008). In LIMA, the dynamical fields in the troposphere and lower stratosphere are relaxed every six hours toward operational reanalyses, primarily ERA-Interim and for historical periods, the NCEP/NCAR 20th Century Reanalysis so that synoptic-scale waves and planetary-wave phases evolve in step with the real atmosphere. Above the nudging domain LIMA integrates the primitive equations freely, resolving the zonally asymmetric mean circulation, tides, and planetary waves that shape the middle atmosphere. Short-wave forcing is driven by daily solar spectra, with explicit absorption of $Ly\alpha$ (Lyman-alpha radiation, 121.6 nm) by O_2 and spectrally integrated heating by O_2 and O_3. Trace-gas climatologies for ozone and carbon dioxide feed a full non-LTE radiative transfer scheme that includes near-IR CO_2 heating, chemical energy storage and airglow loss, plus cooling parameterizations for O_3, H_2O, atomic oxygen, NO_x, and CO_2 across their respective altitude regimes (Kiliani et al., 2013).

The model reproduces the observed mean state of the summer mesosphere, but two deficiencies are well documented. First, the simulated residual circulation is slightly too weak, leading to temperatures in the polar mesopause that are several kelvin warmer than LIDAR climatologies. Second, gravity-wave variability on horizontal scales shorter than about 800 km is under-represented, because sub grid orographic and convective sources are parameterized but not fully resolved.

Grid-point output from LIMA's three hourly fields of temperature, pressure, density, and the full three-dimensional wind vector is interpolated onto the particle ensemble of the *Mesospheric Ice Microphysics and Transport* (MIMAS) model. MIMAS then follows individual meteoric smoke particles as potential ice nuclei, allowing us to examine how the evolving background supplied by LIMA controls nucleation, growth and optical properties of noctilucent clouds.

2.2. MIMAS

The *Mesospheric Ice Microphysics And tranSport* model (MIMAS; Berger & Zahn, 2002) is a three dimensional Lagrangian transport code tailored to noctilucent cloud studies. It follows individual meteoric–smoke particles as potential ice nuclei, advects them with prescribed winds, and updates their radii in response to ambient temperature, water-vapor supersaturation and turbulent diffusion. MIMAS is nested inside the LIMA circulation field and runs only in the region where NLCs form: poleward of 37° N and between 77.8 and 94.1 km altitude. The horizontal grid is 1° in latitude and 3° in longitude; the latter shrinks with latitude according to

$$\Delta s = \frac{3}{360} \cos\left(\frac{\pi\Theta}{180}\right) \times 40\,000 \text{ km}, \tag{2.1}$$

where Δs is the longitudinal grid spacing at latitude Θ, the factor 3/360 converts the 3° resolution into a fraction of the Earth's circumference, $\cos(\Theta)$ accounts for the meridional convergence of longitude lines, and 40 000 km approximates Earth's equatorial circumference. This formulation ensures that at 70° N the longitudinal and latitudinal grid spacings are nearly equal. Vertically, the model resolves 163 layers at 100 m separation, enabling MIMAS to capture the sharp thermal and compositional gradients of the mesopause.

Simulations start on 10 May and end on 31 August, bracketing the full northern summer season. Most physical processes advance with a three-minute time step, while water-vapour advection uses a finer 90-second step to satisfy the courant criterion. Roughly 40 million Lagrangian parcels are tracked; their positions and radii are written once per day, whereas three-dimensional fields of water vapour and derived NLC parameters are archived hourly.

As sites of special interest Kühlungsborn (54°N), ALOMAR (69°N) and Spitsbergen (78°N) particle information is saved at every model step, enabling direct comparison with local lidar and camera observations.

The present work investigates how atmospheric turbulence and vertical wind modify the life cycle of noctilucent cloud particles. We explore these effects numerically by altering the eddy diffusivity and the imposed vertical-wind spectrum within MIMAS. To interpret the resulting sensitivity runs one must first understand how the model transports water vapour and ice particles. Water vapour is treated on the fixed Eulerian grid and advected with a semi–Lagrangian flux scheme, whereas each meteoric-smoke or ice particle is integrated individually in a fully Lagrangian framework that accounts for background wind, turbulent diffusion and gravitational settling. The physical and chemical parameterizations that govern nucleation, growth, sublimation and vapour exchange are described in the next section.

2.2.1. Water vapour distribution

Water vapour is the indispensable precursor for NLCs. Because a full chemistry transport simulation of H_2O production is beyond the scope of MIMAS, the model is started with an empirical source profile $c_0(\varphi, z)$ that depends on latitude φ and altitude z. At the model start date (typically 20 May) the Eulerian water-vapour field is set equal to c_0. During the run it is relaxed back to this profile with Newtonian nudging,

$$\frac{\partial q}{\partial t} = -\frac{q - c_0}{\tau_{\text{relax}}}, \tag{2.2}$$

where q is the mixing ratio and τ_{relax} is 3 days. This weak constraint prevents long-term drift while still allowing transport, diffusion and photolysis to shape the instantaneous distribution. The approach reproduces the seasonal depletion above $\sim$ 84 km and the near-82 km enrichment that are seen in satellite and microwave observations (Berger & Zahn, 2002).

Advection and diffusion
Water vapour is transported on the LIMA wind fields with an explicit flux-form scheme that preserves monotonicity (Bott, 1989; Bott, 1992). In one spatial dimension the advection step solves the continuity equation

$$\frac{\partial q}{\partial t} = -\frac{\partial (u\, q)}{\partial x}, \tag{2.3}$$

where u is the wind component along the axis. Equation (2.3) is integrated every 90 s, satisfying the courant criterion for the 100 m vertical grid spacing used in MIMAS (Kiliani et al., 2013).

Small-scale mixing is represented by a vertical eddy-diffusion coefficient $K_{zz}(z)$ taken from the Andøya rocket climatology of Lübken, 1997 and reduced to 25 % to account for the intermittency of turbulence. Vertical diffusion is implemented with a second-order implicit Crank–Nicolson solver and obeys Fick's second law

$$\frac{\partial q}{\partial t} = K_{zz} \frac{\partial^2 q}{\partial z^2}. \tag{2.4}$$

The scheme remains stable for the 3-min diffusion time-step used, even at the peak value $K_{zz} \approx 45 m^2 s^{-1}$ near 90 km. (Berger & Zahn, 2002)

Equations (2.3) and (2.4) run sequentially within each model step. Together with photodissociation by solar Lyman-α radiation and the nudging term above, they keep the H_2O field in realistic balance while allowing mesospheric processes such as, freeze drying during NLC growth and re-moistening during particle sublimation to emerge naturally.

Photodissociation of H_2O
Water vapour in the summer mesosphere is exposed almost continuously to solar ultraviolet radiation. Photons with wavelengths shorter than about 200 nm carry enough energy to break the H–O bond, producing atomic hydrogen and the hydroxyl radical in a process known as photolytic dissociation. The dominant contributor in the NLC environment is the Lyman–α line at 121.6 nm, which penetrates to roughly 90–95 km before being absorbed; most shorter-wavelength vacuum-UV radiation is removed higher up by O_2 in the Schumann–Runge continuum.

These three processes basically control the spatial and temporal distribution of water vapor inside the MIMAS system.

2.2.2. Multiple–particle system

Ice can form in the summer mesosphere only when the local saturation ratio $S = P_{H_2O}/P_{sat}(T)$ exceeds unity. And it can be through one of these ways: (i) *homogeneous nucleation*, where water molecules cluster spontaneously, and (ii) *heterogeneous nucleation*, where they condense on pre-existing surfaces. Homogeneous nucleation demands very large supersaturation conditions ($S \gtrsim 40$) and temperatures below $\sim$135 K; such conditions occur only sporadically near the very cold summer polar mesopause and over narrow latitude belts(Gadsden, 1990; Gadsden & Taylor, 1994). The more probable pathway is heterogeneous nucleation on meteoric smoke particles (MSPs), microscopic debris that forms when ablated meteoroid material re-condenses between 70 and 100 km altitude (Turco et al., 1982). MIMAS therefore ignores the homogeneous route and treats heterogeneous nucleation as the sole source of NLC ice.

To represent the MSP reservoir the model introduces $\approx 40 \times 10^6$ inert Lagrangian nuclei at the start of each simulation. Particles are seeded uniformly along the 69° N latitude circle (bounding great-circle advection) within an 80–90 km altitude slab and are partitioned into five size classes that follow the observed inverse power-law spectrum. In Berger & Zahn, 2002, they only intorduce 2 million particles and the size classification of those particle is goven in table 2.1, this is scaled to 40 million later by Berger & Lübken, 2015.

Radius Range (nm)	Number of Particles
$1.0 \leq r < 1.5$	1,729,804
$1.5 \leq r < 2.0$	234,000
$2.0 \leq r < 2.5$	31,600
$2.5 \leq r < 3.0$	4,196
$3.0 \leq r < 3.5$	400

Table 2.1: Distribution of particles by radius range (Berger & Zahn, 2002)

Within each class the initial radius is assigned from a uniform random distribution. For a given ambient supersaturation, the Kelvin equation yields a minimum nucleus radius

$$r_{\text{min}} = \frac{2\sigma_{\text{ice}}}{\rho_{\text{ice}} RT \ln S}, \tag{2.5}$$

below which particles cannot grow by condensation.

Here, r_{min} is the Kelvin (critical) radius for activation, σ_{ice} is the surface tension of the ice–vapour interface, ρ_{ice} is the bulk density of ice, R is the universal gas constant, T is the ambient temperature, and S is the water-vapour supersaturation ratio. Particles with $r < r_{\text{min}}$ tend to evaporate, whereas particles with $r > r_{\text{min}}$ can grow by vapour deposition.

At Kühlungsborn's latitude (54° N) this threshold is $r_{\text{min}} \approx 0.7$ nm for $S \approx 70$; larger nuclei therefore activate sooner and at lower supersaturation than smaller ones. In practice class-4 particles ($r > 3$ nm) can begin ice growth at $S \gtrsim 4$ and altitudes $\gtrsim$ 84 km, whereas class-1 particles ($r < 1.5$ nm) require $S \gtrsim 10$. Once nucleated, each particle is tracked individually advected by the LIMA wind, mixed by turbulent diffusion and allowed to grow, sublimate and sediment, so that the ensemble evolves self consistently into the complex, multi-modal ice population characteristic of noctilucent clouds.

2.2.3. Lagrangian transport scheme of particles

The motion of each particle is followed individually on the three–dimensional grid. For a given time step Δt, the position of particle n at time index i evolves according to

$$\begin{aligned} x_{i+1}^n &= x_i^n + u_i^n \,\Delta t, \\ y_{i+1}^n &= y_i^n + v_i^n \,\Delta t, \\ z_{i+1}^n &= z_i^n + w_i^n \,\Delta t, \end{aligned} \tag{2.6}$$

where the superscript n denotes the particle index and the subscript i denotes the discrete time step. The quantities u_i^n and v_i^n are the zonal and meridional wind components interpolated from the LIMA background field, such that $u_i^n = u_{\text{b}i}^n$ and $v_i^n = v_{\text{b}i}^n$. The vertical velocity is composed of three contributions,

$$w_i^n = w_{\text{b}i}^n + w_{\text{s}i}^n + w_{\text{e}i}^n,$$

where $w_{\text{b}i}^n$ is the background vertical wind, $w_{\text{s}i}^n$ is the sedimentation velocity of the particle, and $w_{\text{e}i}^n$ represents the turbulent (eddy-induced) vertical motion. A schematic illustration of the forces controlling particle trajectories is shown in Figure 2.2.

The sedimentation velocity depends on particle radius and ambient conditions:

$$w_s = -\frac{g\rho r}{2n_{atm}}\sqrt{\frac{\pi}{2m_{atm}k_B T}} \tag{2.7}$$

where $g = 9.55\ \text{m}\,\text{s}^{-2}$ is gravitational acceleration, $m_{\text{atm}} = 4.845 \times 10^{-26}$ kg the mean molecular mass of air, n_{atm} the air number density, ρ the bulk ice density, and T the local temperature. This equation is adapted from Chapman & Kendall, 1966 and Reid, 1990. On the other hand, turbulent diffusion is parameterised by a random vertical velocity

$$w_{\text{e}i}^n = P_r\, w_{\text{e}}(z), \qquad P_r \in [-1, 1], \tag{2.8}$$

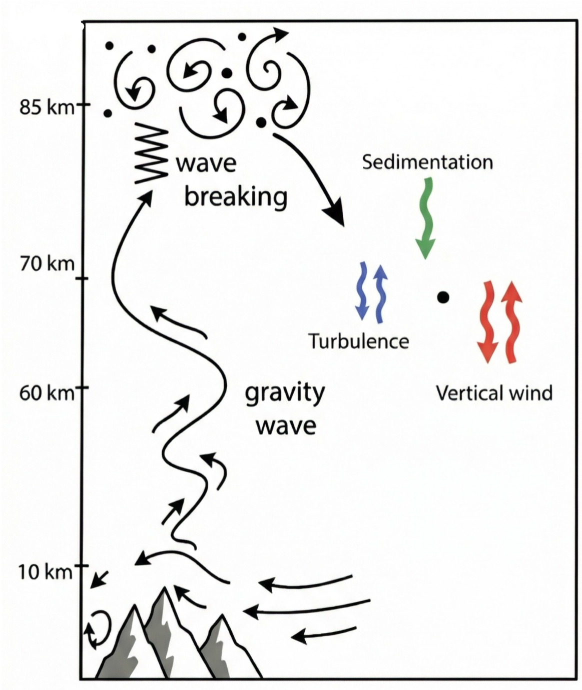


Figure 2.2: Schematic representation of the forces and transport processes acting on an individual ice particle in MIMAS. The diagram illustrates the contributions from the synoptic background vertical wind w_{bi} , gravitational sedimentation w_{si} , and turbulent eddy motions w_{ei}, which together determine the total vertical velocity of a particle.

where P_r is a uniformly distributed random number. This construction ensures that, in an ensemble sense, the particles reproduce Fickian diffusion with coefficient $K_{zz}(z)$, because the mean square displacement over a large sample satisfies $\langle \Delta z^2 \rangle = 2K_{zz}\,\Delta t$ (Berger & Zahn, 2002).

The random turbulent "kicks" are chosen so that their statistical properties reproduce Fickian diffusion; in other words, the ensemble of particles follows the analytic solution of the diffusion equation without introducing artificial numerical damping. Because the half-width of a diffusing particle layer increases in proportion to $\sqrt{K_{zz}}$, changing K_{zz} in the sensitivity experiments directly controls how far the cloud spreads, exactly as predicted by diffusion theory.

2.2.4. Experimental set-up

With the transport scheme established, we turn to the sensitivity experiments designed to isolate the roles of turbulence and vertical wind. Ideally one would vary these fields by adjusting the gravity wave parameterization inside LIMA, but that scheme resolves only waves with horizontal wavelengths longer than about 800 km and therefore under-represents the small-scale variability that drives both eddy diffusion and mesospheric up- and downdrafts. To compensate for this, the turbulence and vertical wind are modified separately.

Vertical wind

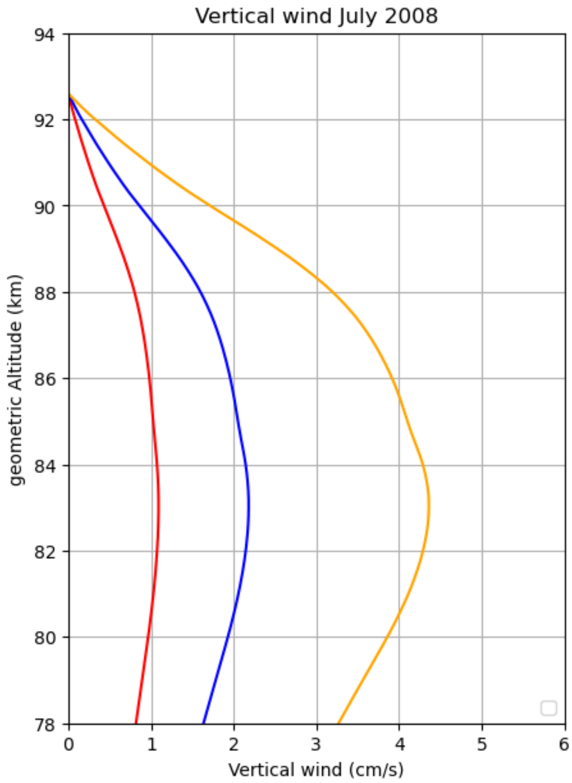


Figure 2.3: altitude profile of Vertical wind profile used as background (w_{bi}) for each experiment. Blue represents the reference vertical wind, red the reduced (50%) and yellow represents the increased levels (200%).

The vertical wind obtained from LIMA is scaled to conduct the experiments. In Fig. 2.3, the middle (blue) line represents the reference vertical wind, i.e. the unmodified LIMA output. Two additional scenarios are created by using 50% of this wind (red) and 200% of the reference (orange). Note that during summer the mean vertical wind in the domain is positive, i.e. predominantly upward.

Turbulence

The turbulence (eddy diffusivity K_{zz}) is originally based on Lübken's rocket measurements from 1997 (Lübken, 1997). Following Berger & Zahn, 2002, only 25% of those observed values are used as the MIMAS reference, as this choice yielded more realistic vertical structure. In the present experiments we investigate what happens when the vertical turbulence is varied around that reference. Specifically, we adopt the Berger & Zahn, 2002 profile as 1.0×, and we also test 0.5× (50% of reference) and 2× . The K_{zz} profiles used are shown in Figure 2.4.

Even though many experiments were conducted to confirm the results and address the main scientific questions that arose during the project, most of these results are not included in the report. Some have been provided in the supplementary material for the reader's reference.

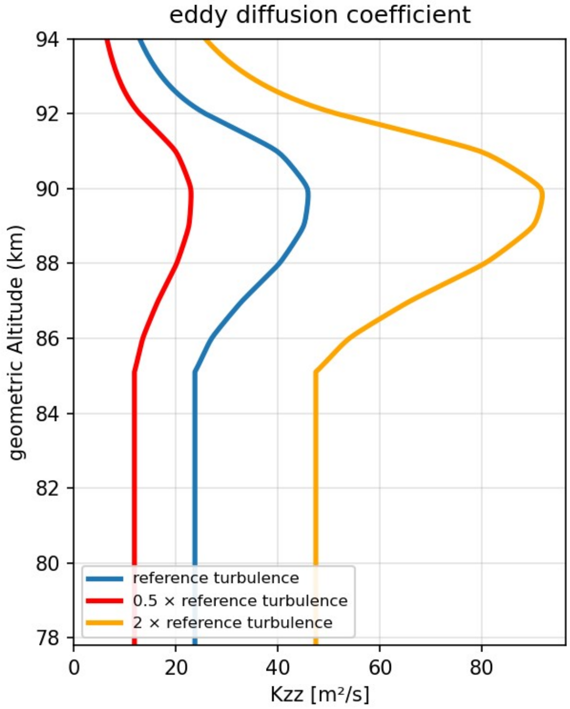


Figure 2.4: altitude profile of turbulence (eddy diffusion coeffficient, k_{zz}) for each experiment. Blue represents the reference vertical wind, red the reduced (50%) and yellow represents the increased levels (200%).

3

Results and Discussion

3.1. Effect of Turbulence

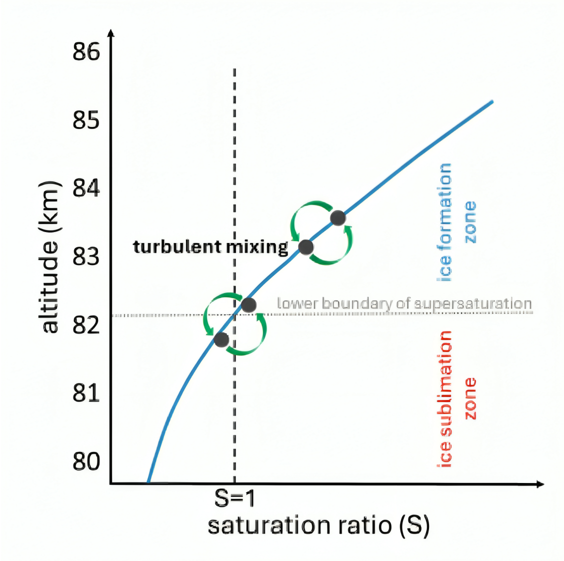
3.1.1. Effects on saturation ratio

Understanding the saturation ratio of water vapour in this region is very important to predict and assess the behaviour of cloud formation. As discussed earlier, since the water vapour concentration is enough and the temperature is very low, higher saturation ratios make it feasible for condensation nuclei to form ice particles. At the same time, very high supersaturation values are required to initiate heterogeneous nucleation. In MIMAS, the degree of saturation water vapour is calculated as:

$$S = P_{H_2O}/P_{ice} \tag{3.1}$$

Here, P_{H_2O} is the partial pressure of water vapour, given by $C_{H_2O} \cdot P_{atm}$ (where C_{H_2O} is the water vapour mixing ratio), and P_{ice} is the saturation vapour pressure over a flat ice surface. In MIMAS, P_{ice} is calculated using the following formula, which is obtained from Berger & Zahn, 2002:

$$P_{ice} = \exp\left(9.5504 - (5732.265/T) + 3.5306 \times \log_{10}(T) - 0.00728322 \times T\right) \tag{3.2}$$



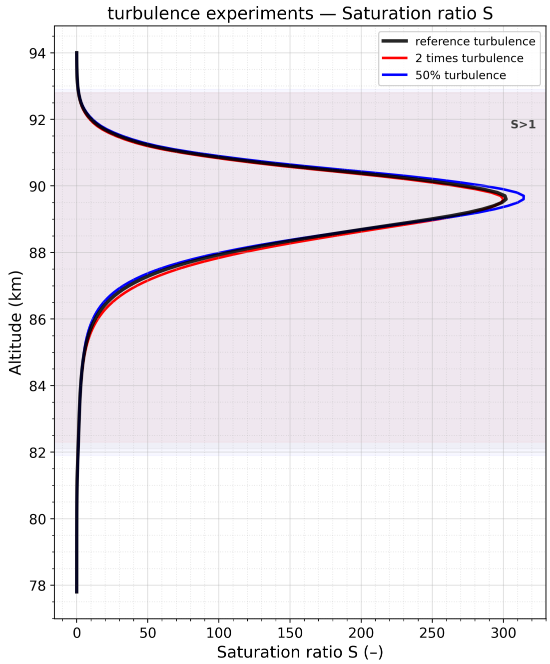


Figure 3.1: On the left: Schematic illustration of the saturation ratio profile and the growth–sublimation boundary. The blue curve indicates the height dependence of the saturation ratio S, with the region where S>1 representing the ice-formation zone. Arrows indicate turbulent mixing transporting particles across the S=1 threshold. Figure adapted from , manuscript submitted to Frontiers in Astronomy and Space Sciences (Vellalassery et al., 2025). **on the right:** Saturation ratio calculated using equation 3.1 for all three turbulence cases with ice formation (Averaged for the month of July over $69°N$). The shaded region in the background indicates the 'ice formation zone', i.e., $S > 1$.

When $S > 1$, the environment is supersaturated, meaning ice particles can grow, while when $S < 1$, the environment becomes subsaturated, leading to the sublimation of ice particles. The value of S for different turbulence conditions is calculated and shown in Figure 3.1. In this thesis, the region where $S > 1$ is defined as the *"ice formation zone"*, and the region below this as the *"ice sublimation zone"*. These terms will be used throughout the report to describe the respective regions. The ice formation zone is shaded in every vertical profile plot for the reader's convenience. The concept of ice formation zone is visualised in figure 3.1.

The saturation ratio required for heterogeneous nucleation varies with particle radius, smaller particles need significantly higher S values to act as condensation nuclei. S values reach their highest near the mesopause region (around 88 km), where the temperature is the lowest. Even the smallest dust particles (represented in MIMAS) can start to accumulate water vapour and form ice particles around this region, which then sediment downward due to gravitational pull. It is also interesting to note that the local S values show only limited variation when the turbulence is changed. As temperature and water vapour are the most significant factors influencing S, it is reasonable to assume that the water vapour within the column is not strongly disturbed by fluctuations in turbulence.

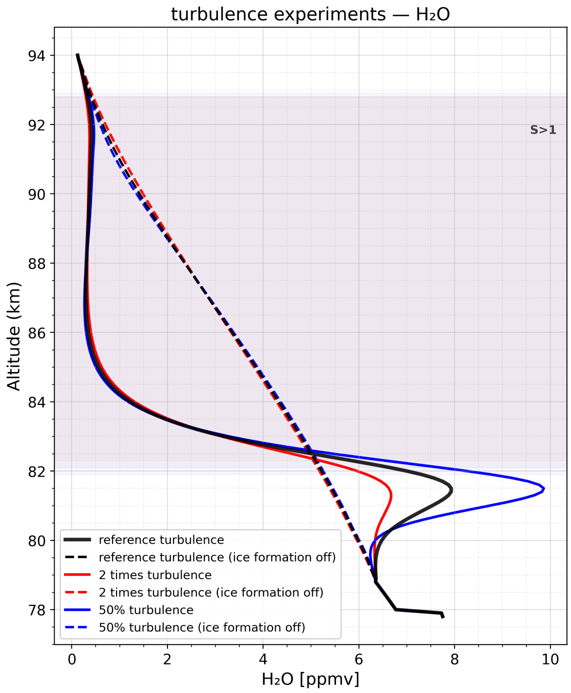


Figure 3.2: July–mean water vapour profiles under different turbulent conditions at 69°N.

3.1.2. Impact of turbulence on water vapour

To analyses the impact of turbulence on water vapour in more detail, the water vapour concentration in the atmosphere is examined under different turbulence conditions. Figure 3.2 shows the zonally and July averaged water vapour (H_2O) profiles from MIMAS for three different turbulence scenarios, both with and without enabling ice formation in the model.

A clear difference in the water vapour distribution can be seen between the simulations with NLC formation (solid lines) and without NLC formation (dashed lines). In a normal atmosphere, without ice formation, the water vapour concentration decreases from around 10 ppmv to less than 1 ppmv between 80 and 90 km altitude. When turbulence is varied, no major change is observed in this pattern. However, once ice formation is enabled, the background water vapour drops drastically. This is because nucleation particles absorb water vapour from the surrounding atmosphere under favourable conditions in order to grow, leading to a noticeable reduction in the background water vapour concentration once ice formation begins.

Even when the turbulence is doubled, the water vapour concentration does not fluctuate much. This is due to the relatively weak role of turbulent eddy diffusion in vertical water transport at mesospheric altitudes. The influence of turbulence on water vapour concentration can be expressed by the following equation:

$$\frac{\partial H_2O}{\partial t} = K_{zz}(z) \cdot \frac{\partial^2 H_2O}{\partial z^2} \tag{3.3}$$

Here, K_{zz} is the height-dependent eddy diffusion coefficient obtained from Lübken, 1997. Since the absolute concentration of water vapour in this region is extremely low, even a large change in turbulence strength does not significantly affect the vertical distribution. On the other hand, once nucleation particles are available, the resulting ice particles are treated as individual entities and transported using the particle transport scheme (see Chapter 2.2.3). Because the random motions of particles in MIMAS are directly proportional to the K_{zz} values, the effect of turbulence becomes clearly visible once ice formation is activated.

In the ice formation zone, a significant depletion in water vapour concentration is observed. This occurs because ice particles continuously absorb water vapour from the background air to grow larger under suitable conditions. Many ice particles are created near the mesopause region and then travel downward due to gravitational pull a process defined by their sedimentation velocity. The sedimentation velocity at these heights typically ranges between 10 and 100 cm s^{-1} according to Kiliani et al., 2013. As these particles descend, they enter regions where the availability of water vapour is higher, allowing them to continue growing in size. However, once the ice particles reach the lower edge of the ice formation region, the situation changes. The air here is no longer supersaturated, and the ice particles begin to lose mass through sublimation. This process is known as the *freeze drying effect*. The freeze drying effect is responsible for the notable enhancement in H_2O concentration just below the $S = 1$ boundary. As the ice particles sublimate, the released water vapour enriches the local air, producing a sharp increase in H_2O concentration in the sublimation zone.

This behaviour can be clearly observed in Figure 3.3, where the temporal evolution of water vapour over the entire summer season is shown as a function of height. When ice formation is turned off (top row), the water vapour profiles remain smooth and uniform, with no distinct vertical structure or depletion layers visible. The concentration gradually decreases with altitude but remains largely unaffected by the changes in turbulence, confirming that eddy diffusion alone does not cause any significant variability. In contrast, when ice formation is enabled (bottom row), strong vertical gradients appear between 82 and 86 km, the main NLC region. These plots show a clear depletion band corresponding to the ice formation zone and a bright enhancement just below it, representing the sublimation region. The strength of these features varies slightly with turbulence, with the depletion and recovery patterns following the seasonal cycle, strongest in mid-June to mid-July and weakening toward the end of August.

With NLC formation enabled, the effect of turbulence becomes more pronounced in the background water vapour profiles. The hydration at sublimating altitudes (resulting from the freeze drying effect) decreases with increasing turbulence. In other words, the amplitude of water vapour enhancement in the sublimation zone (below roughly 82 km) becomes smaller when turbulence is stronger. As discussed earlier, turbulence induces vertical mixing that moves particles up and down over short distances. When such mixing occurs well within the ice formation zone (above about 84 km), particles remain inside the water vapour supersaturated region. In this case, stronger turbulence enhances particle motion but does not significantly alter their growth rates, since they stay within $S > 1$. In contrast, near the $S = 1$ boundary (around 82 km), increased turbulence pushes particles in and out of the saturation zone, making them repeatedly cross between $S > 1$ (growth) and $S < 1$ (sublimation). This constant exchange plays an important role in defining the microphysical evolution of ice particles in this region. Even small changes in turbulence around the $S = 1$ threshold can therefore lead to noticeable differences in the growth and lifetime of ice particles.

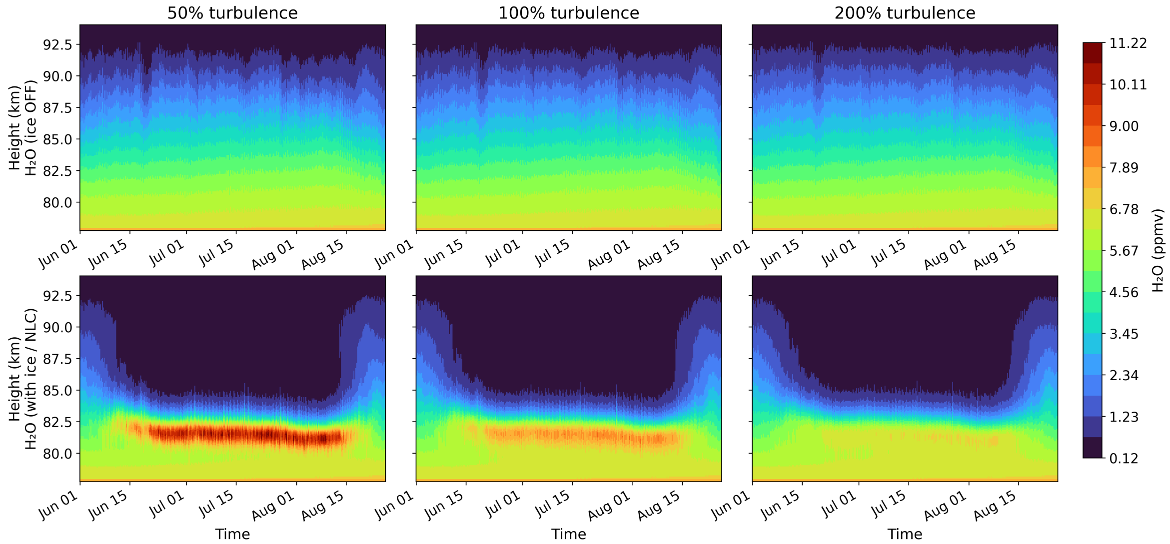


Figure 3.3: Spatial and temporal evolution of water vapour (H_2O) for three turbulence strengths during the summer season over ALOMAR (69°N). The top panels show simulations with ice formation disabled (background transport only), while the bottom panels show runs with ice formation enabled.

This behaviour is further influenced by the Kelvin effect, which describes the dependence of the equilibrium vapour pressure on particle size. For very small ice particles, surface tension creates an internal Laplace pressure that increases the local vapour pressure at the particle surface. As a result, smaller particles require higher saturation ratios to remain in equilibrium with their surroundings. This means that once a particle becomes smaller through sublimation, it needs an even higher S value to start growing again.

For example, turbulence can move particles vertically much faster (on the order of m s^{-1}) than their natural sedimentation velocity of about 10 cm s^{-1} for a 50 nm particle at 83 km altitude (Kiliani et al., 2013). This rapid downward transport can carry particles into subsaturated regions ($S < 1$), where they quickly shrink or disappear due to sublimation. If these smaller particles are later transported back upward, they may not regrow, even if the local S value is slightly above one. Because of their reduced size, the Kelvin effect now demands a higher S threshold for growth than the ambient atmosphere can provide. As a result, such particles continue to sublimate rather than reform, reducing the effective particle number density in this region.

To better understand the characteristics of NLC formation under different turbulence conditions, it is necessary to examine the main NLC micro-physical properties, such as the ice particle number density, mean particle radius, and optical brightness, which are discussed in the following sections.

3.1.3. Impact of turbulence on NLC properties.

The number density of ice particles (n_{ice}) is shown in Figure 3.4 (on the left). A dust particle is considered an ice particle as soon as it begins to accumulate water molecules around it, and the total number of such particles per cubic centimeter is represented by n_{ice}. The number density consistently peaks around 87 km, which corresponds to the mesopause altitude. As seen previously in Figure 3.1, the temperature reaches its minimum at the mesopause, and the saturation ratio (S) reaches its maximum. These very high S values create extremely favorable conditions for even the smallest dust particles in MIMAS to act as condensation nuclei. The only remaining limitation is the availability of water vapour.

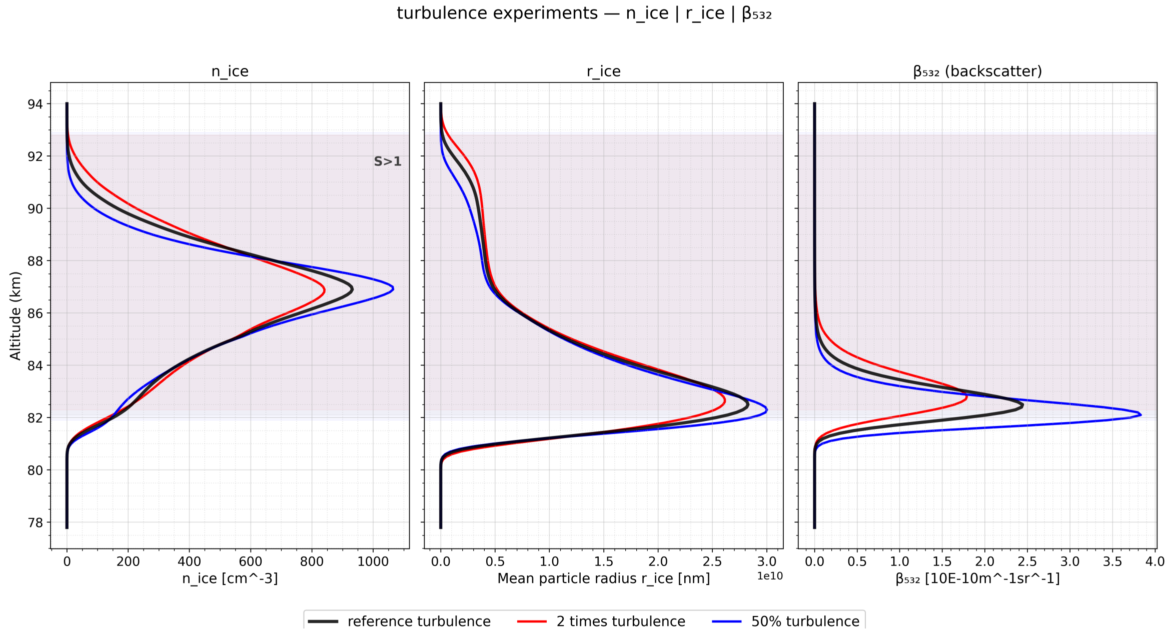


Figure 3.4: Vertical profiles of ice particle number density (n_{ice}), mean particle radius (r_{ice}), and optical backscatter coefficient (β_{532}) for three turbulence conditions.

According to the particle size distribution (Table 2.1), most of the meteoric dust particles are extremely small, about 86.7% of them have radii between 1.2 and 1.7 nm. These smaller particles require very high supersaturation levels to initiate condensation. At the mesopause, where S is highest, these particles finally encounter suitable conditions to begin ice formation. Consequently, the mesopause acts as the primary region of new ice nucleation in the model.

The region where S value reaches really high values is very narrow in vertical extent. When turbulence increases, ice particles and condensation nuclei are pushed in and out of this narrow layer more frequently. As a result, the effective residence time of particles within the highly supersaturated region decreases, leading to a reduction in n_{ice} with increasing turbulence. Conversely, under low-turbulence conditions, the particles remain within this favorable zone for longer, and the number density increases accordingly.

At the same time, turbulence also causes vertical redistribution of the dust particles themselves. In MIMAS, dust particles are initially concentrated near the mesopause, similar to the natural distribution described by Hunten et al., 1980. When turbulence intensifies, these particles are mixed over a broader altitude range, resulting in a slightly higher ice number density at upper altitudes and a smoother vertical profile overall. As altitude decreases, the saturation ratio S declines due to rising temperature, which significantly slows the formation of new ice particles. This explains why n_{ice} peaks around 87 km and gradually decreases toward lower altitudes.

At typical NLC altitudes around 83 km, the modeled particle number densities range from 50 to 200 cm^{-3}, which agrees reasonably well with lidar observations reported by Baumgarten et al., 2008. Overall, these results indicate that turbulence primarily influences the vertical spread and local concentration of ice particles rather than their total number. Stronger turbulence broadens and flattens the vertical distribution of n_{ice}, while weaker turbulence allows a sharper peak centered at the mesopause. This demonstrates that during the early stages of ice particle growth, vertical motion in the NLC region is dominated by turbulence, leading to a more extended and diffused ice layer at higher K_{zz} values.

Ice particles, once formed, are subjected to motion caused by background wind (w_{bi}), turbulent diffusion (w_{ei}), and gravitational settling (w_{si}). These processes transport particles into different regions of the atmosphere, where they experience new temperature and water vapour conditions. The combination of their size, ambient temperature, and local water vapour availability determines the fate of each ice particle. Regardless of atmospheric conditions, all particles gradually sediment to lower altitudes, where temperatures and saturation ratios (S) are lower, but water vapour concentration is comparatively higher. If an ice crystal is large enough to allow continued condensation under these lower S conditions, it will continue to grow in size. Hence, analysing the mean radius of ice particles (r_{ice}, Figure 3.5 middle panel) provides important insight into their sedimentation behaviour and how it responds to changes in turbulence.

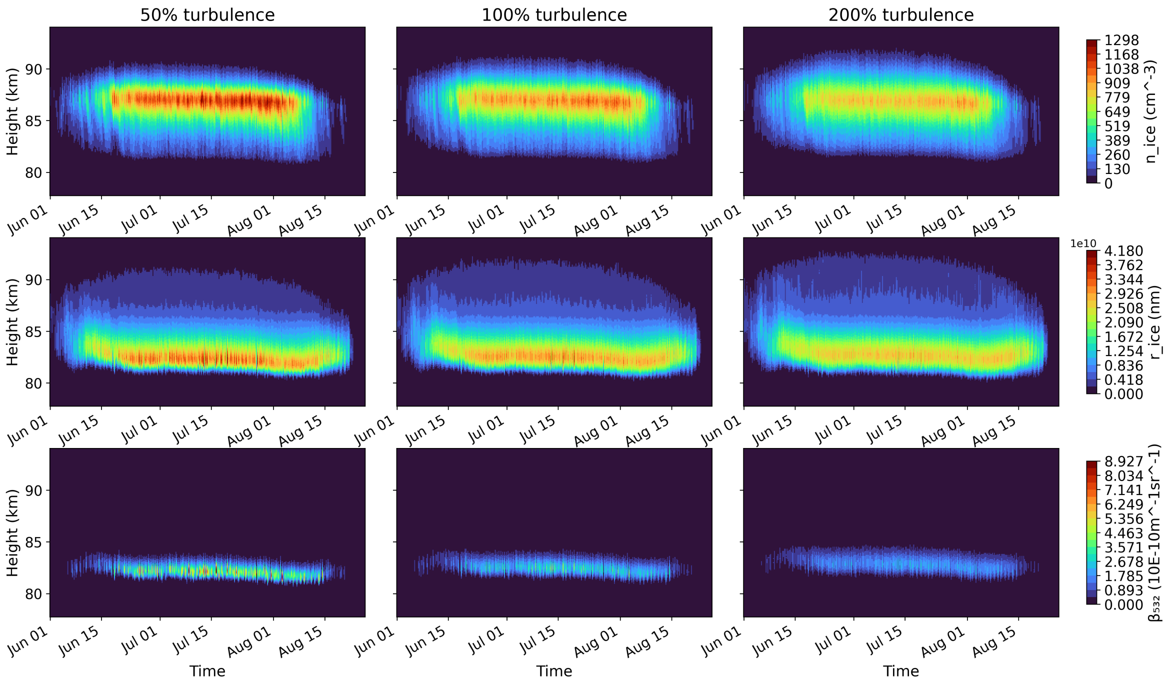


Figure 3.5: Spatiotemporal variation of key NLC properties under different turbulence conditions. Columns represent turbulence values of 50%, 100%, and 200% relative to the reference case. Rows show, from top to bottom, the number density of ice particles (n_{ice} [cm^{-3}]), mean particle radius (r_{ice} [nm]), and optical backscatter coefficient (β_{532} [10E^{-10} m^{-1}sr^{-1}]).

Comparing different turbulence scenarios shows that an increase in turbulence leads to a decrease in the maximum ice particle radius (see figure 3.5), while the altitude of large-sized particles shifts slightly upward. As previously discussed, most ice particles are small and form near the coldest region close to the mesopause (86–88 km), from where they settle downward.

During this descent, they are affected by mean winds, wave motions, and turbulence. Not all particles that form near the mesopause grow efficiently, smaller ice particles sublimate quickly due to the Kelvin effect, releasing their water vapour as they descend into warmer, subsaturated regions. This released vapour then becomes available for condensation on slightly larger, more stable particles, enhancing their growth.

Although S decreases with decreasing altitude due to rising temperature, the increased availability of water vapour from sublimating small particles promotes further growth of larger ones. Ice particles typically reach their maximum size just above the sublimation zone, around 83–84 km, where temperatures begin to exceed the frost point. More than 80 % of the total particle growth occurs within this narrow region over a short period of time (Berger & Zahn, 2002). The net sedimentation speed of ice particles and their residence time within this maximum growth zone are therefore key parameters controlling NLC properties. Increased turbulence near the $S = 1$ boundary enhances particle movement across the growth–sublimation interface, accelerating sublimation and reducing particle size, thereby shortening their residence time in the region of most efficient growth. This explains the observed decrease in maximum ice particle radius with increasing turbulence and the slight upward shift in the altitude of the largest particles.

We have also conducted a comprehensive analysis of the variable β_{532}, which represents the backscatter value for the wavelength 532 nm. This variable is very important, as most NLC observations are made using LIDARs located at various sites on Earth, including ALOMAR and IAP. Therefore, to compare the model results with observations, β_{532} or brightness (referred to in this thesis as 'NLC brightness') data is necessary.

Figures 3.5 illustrate the time series of NLC brightness under different turbulence conditions (bottom row). Although the maximum number of ice particles forms at higher altitudes, the altitude of maximum NLC brightness is around 82–83 km. This is because the radius of the ice particles primarily determines the brightness of NLCs; therefore, the maximum brightness corresponds to the altitude where the radius is largest, just above the sublimation zone (Kiliani et al., 2013). This is consistent with previous observational studies, which show that brighter NLCs are more strongly associated with larger particle sizes than with higher number densities (Baumgarten et al., 2008; Rapp & Thomas, 2006).

When the turbulence increases, particles struggle to grow larger. As a result, the brightness decreases significantly. Additionally, the altitude of maximum brightness shifts higher, and the brightness distribution broadens due to turbulence-induced dispersion. NLC backscatter strongly depends on particle radius (r). For particles smaller than 30 nm, Rayleigh scattering dominates ($\beta \propto r^1$), while for particles around 30–50 nm, Mie scattering applies, where ($\beta \propto r^6$), making the backscatter extremely sensitive to changes in particle size ((Hervig et al., 2009)). Therefore, even small changes in radius lead to significant changes in optical scattering. Doubling the reference turbulence causes an approximate 7% reduction in the peak mean ice particle radius, leading to a 30% decrease in the maximum mean brightness.

Most of the primary observations suggest that turbulence affects the NLC cloud and its properties differently at different heights. This is due to the uneven distribution of H_2O, dust particles, and temperature profiles at different altitudes. To investigate this further and to bring clarity to our findings, more experiments were conducted.

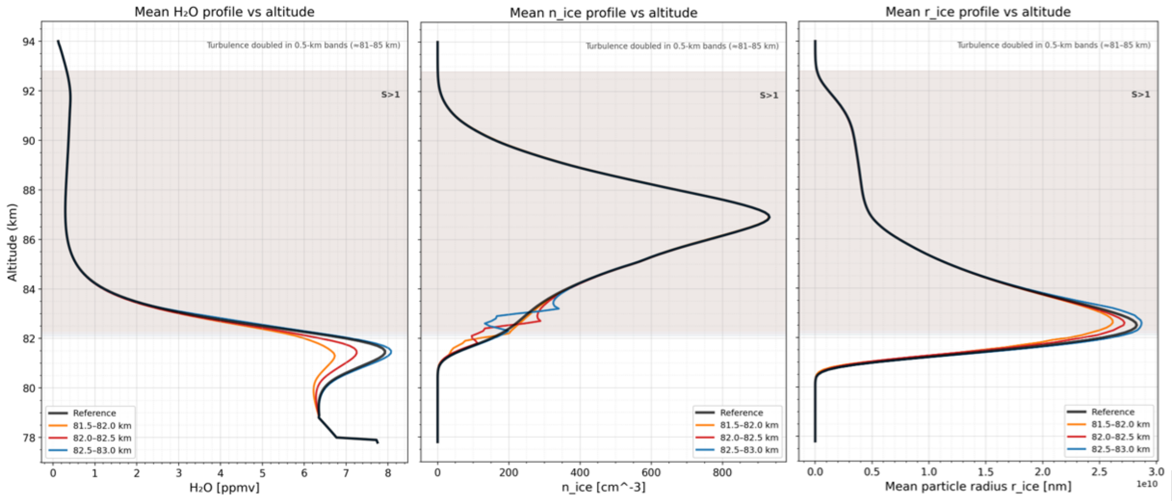


Figure 3.6: Sensitivity of NLC microphysical parameters to turbulence at different altitudes. The three panels show vertical profiles of (from left to right) H_2O [ppmv], n_{ice} [cm^{-3}], and mean particle radius r_{ice} [nm] under reference and locally enhanced turbulence conditions. Turbulence is doubled in 0.5 km-wide layers centered at 81.5–82.0 km (blue), 82.0–82.5 km (orange), and 82.5–83.0 km (red).

3.1.4. Altitude dependency of Turbulence

To investigate the altitude dependency of turbulence over NLC clouds, we conducted a few experiments by doubling the turbulence value in 500 m layers, starting from 80 to 85 km. This region is selected because most NLCs occur within this altitude range. Here we focus on three specific experiments where turbulence is enhanced between 81.5–82.0 km, 82.0–82.5 km, and 82.5–83.0 km. These layers were selected because the $S = 1$ boundary exists roughly between 81 and 83 km. As discussed earlier, if the strong up-and-down motion caused by turbulence pushing particles in and out of the ice formation zone is the main driver of variability, we should be able to observe this effect clearly here.

The results of these experiments are shown in Figure 3.6. When turbulence is increased below the $S = 1$ line, ice formation decreases, showing a reduction in the freeze drying effect, smaller particle radii, and limited ice growth. In contrast, when turbulence is increased within the ice formation zone itself, the NLCs exhibit the opposite effect the average maximum radius increases, and the freeze drying effect becomes more pronounced. When turbulence is enhanced well inside the ice formation zone, it loses its ability to push ice particles out of this region but instead causes them to shuttle back and forth within it. This results in a slight positive impact, as it increases the mean residence time of large ice particles typically observed around these altitudes. As a result a slight increase int he radius of ice particle occurs (figure 3.6).

At the lower part of the ice formation region, any downward turbulent motion can push particles into warmer, subsaturated conditions where they can no longer exist as ice. This leads to a decrease in n_{ice}. This confirms the earlier results. It can be seen that when turbulence is suddenly increased in 500 m patches, the number of particles shows a clear increase just above the top of the patch, while within and below the patch itself, n_{ice} decreases compared to the reference case.

Overall, these sensitivity experiments demonstrate that the effect of turbulence on NLCs is highly altitude dependent. Turbulence introduced near the mesopause or within the upper part of the ice formation zone tends to redistribute particles without removing them from supersaturated conditions, whereas turbulence introduced near the lower growth sublimation boundary has a strongly erosive effect on the cloud. In the real atmosphere, such altitude dependent variability may arise from intermittent gravity-wave breaking, which produces localized bursts of turbulent energy at different heights. Depending on where and when these events occur, they can imprint distinct signatures on NLC number density, particle size, and vertical structure.

3.2. Effect of Vertical Wind

Winds in the upper atmosphere are generally weak, typically only a few centimeters per second, but they fluctuate strongly with season, latitude, and gravity wave activity. Even though these values look small, vertical wind plays a very direct role in controlling both the sedimentation of ice particles and the transport of water vapour into the NLC region. In this section, we investigate how changes in vertical wind influence the formation of NLCs and their background environment.

3.2.1. Impact of vertical wind on water vapour

At summer–mesosphere altitudes (80–90 km), the climatological vertical wind is predominantly upward, driven by gravity wave breaking and the residual meridional circulation (Lindzen, 1981; Garcia & Solomon, 1985). Even an upward motion of a few $\mathrm{cm\,s^{-1}}$ can slow down the sedimentation of ice particles and at the same time transport additional water vapour from lower altitudes into the NLC formation region.

Figure 3.7 shows the temporal evolution of vertical wind during July, averaged over 69°N. The pattern clearly shows that the vertical wind remains upward through most of the month, but with strong fluctuations on daily timescales. These oscillations can momentarily strengthen or weaken the upward transport of water vapour, causing subtle variations in the moisture supply to the ice forming zone.

Gravity wave breaking in the summer mesosphere (around 80–88 km) deposits momentum that forces strong upward motion, creating the well-known upwelling and cooling in this region. But, near and above the mesopause (above 90 km), the pattern of momentum deposition changes because many gravity waves are filtered out, and strong solar heating in the lower thermosphere becomes dominant. This combined effect reverses the circulation and produces downward motion above 90 km. So the lower layers show upward winds due to gravity-wave breaking, while the upper layers show downward winds because the drag weakens or reverses and thermospheric heating drives sinking.

Figure 3.8 shows the July mean water vapour profiles for all three vertical wind experiments. In contrast to turbulence, which redistributes ice particles up and down and significantly alters their number density changes in vertical wind mainly shift the entire H_2O profile slightly upward, without strongly modifying its shape or magnitude. This behaviour highlights the difference between the two processes: turbulence drives chaotic small scale mixing, whereas vertical wind acts more like a largescale “conveyor belt” that slowly pushes the background air upward.

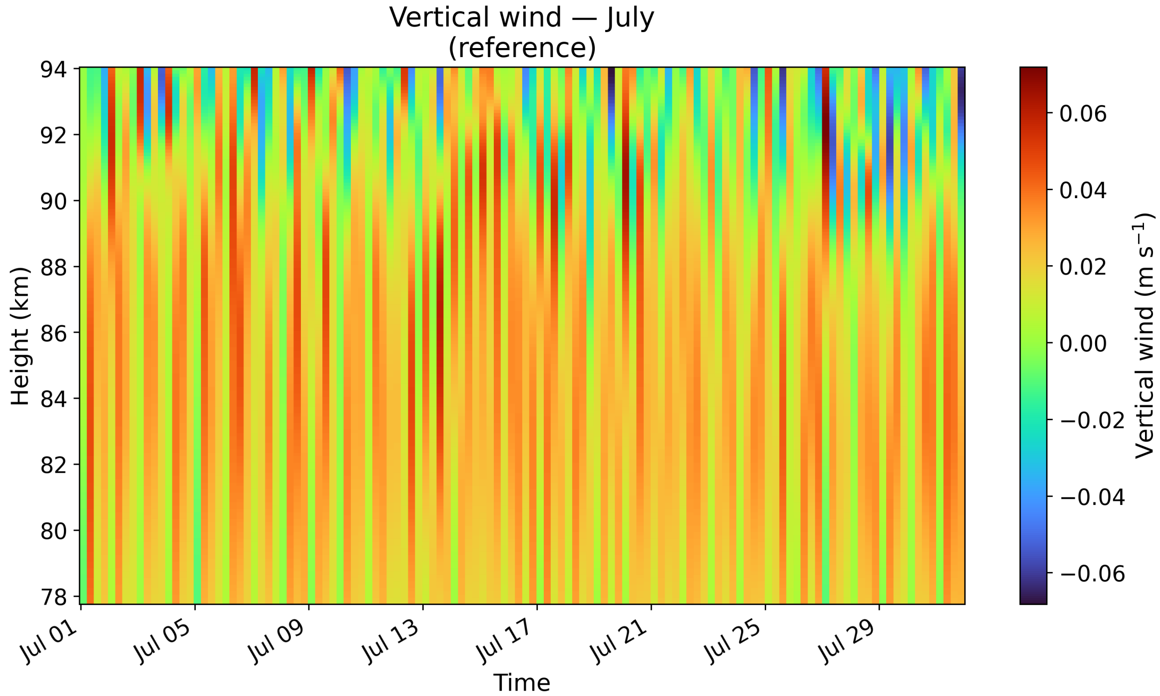


Figure 3.7: Temporal evolution of vertical wind during July (reference run), averaged over 69°N. Positive values indicate upward motion.

Because MIMAS uses prescribed temperatures from LIMA, changes in S originate entirely from the modified water vapour field. When vertical wind is increased, more water vapour is transported upward, causing the lower boundary of the $S = 1$ line to shift downward, while the upper boundary moves slightly downward in altitude due to the slightly high downward push from the upper layers.

The spatial plots in Figure 3.9 show how the background water vapour field evolves through the summer months when the vertical wind is changed. One clear difference from the turbulence experiment is that vertical wind affects the entire H_2O column rather than just the narrow ice–formation region. When the wind is weak (50%), the H_2O field looks smooth and almost steady. As the vertical wind increases, the water vapour becomes much more varying in time, and the day-to-day fluctuations become noticeably stronger. At the same time, the altitude of maximum water vapour gradually moves upward. This upward shift is small in the 100% case but becomes very clear when the vertical wind is doubled, where the moist layer rises by roughly 1–2 km and also appears earlier in the season.

Another interesting feature is the way the freeze drying region reacts when vertical wind increases. The bottom row of Figure 3.9 shows that once ice formation is allowed, the increase in water vapour below the sublimation zone (around 81–83 km) becomes much stronger than expected. In other words, the interaction between vertical wind and the freeze drying effect is not linear: doubling the wind produces more than double the increase in water vapour in this region. This also changes the overall seasonal evolution of the ice formation environment. With stronger upward wind, the humid layer below 83 km becomes deeper and more persistent, while the upper part of the H_2O profile above 88 km becomes smoother and more extended. Together, these changes show that vertical wind does not simply “shift”

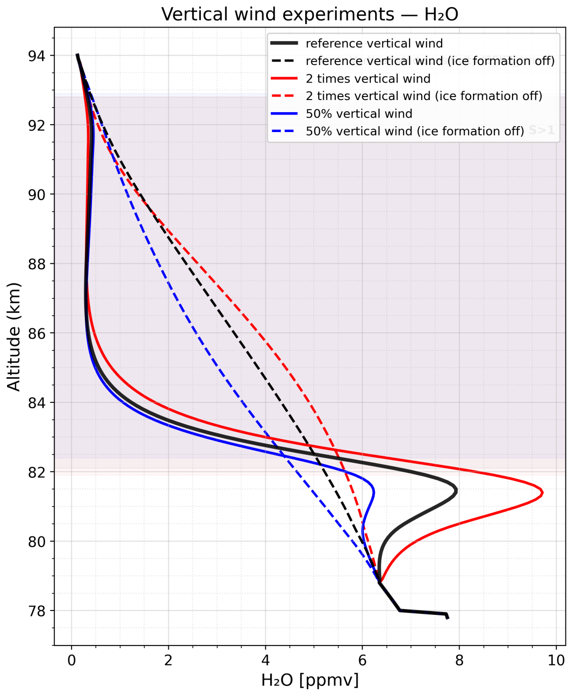


Figure 3.8: July–mean water vapour profiles under different vertical–wind conditions at 69°N.

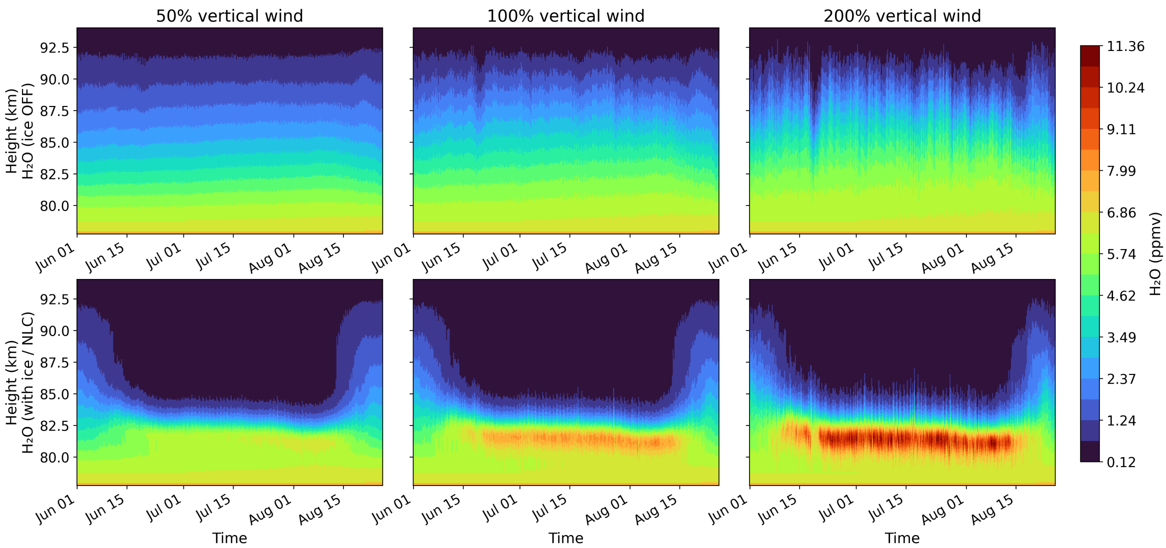


Figure 3.9: Spatial and temporal evolution of water vapour (H_2O) for three vertical wind strengths during the summer season over ALOMAR ($69°N$). The top panels show simulations with ice formation disabled (background transport only), while the bottom panels show runs with ice formation enabled.

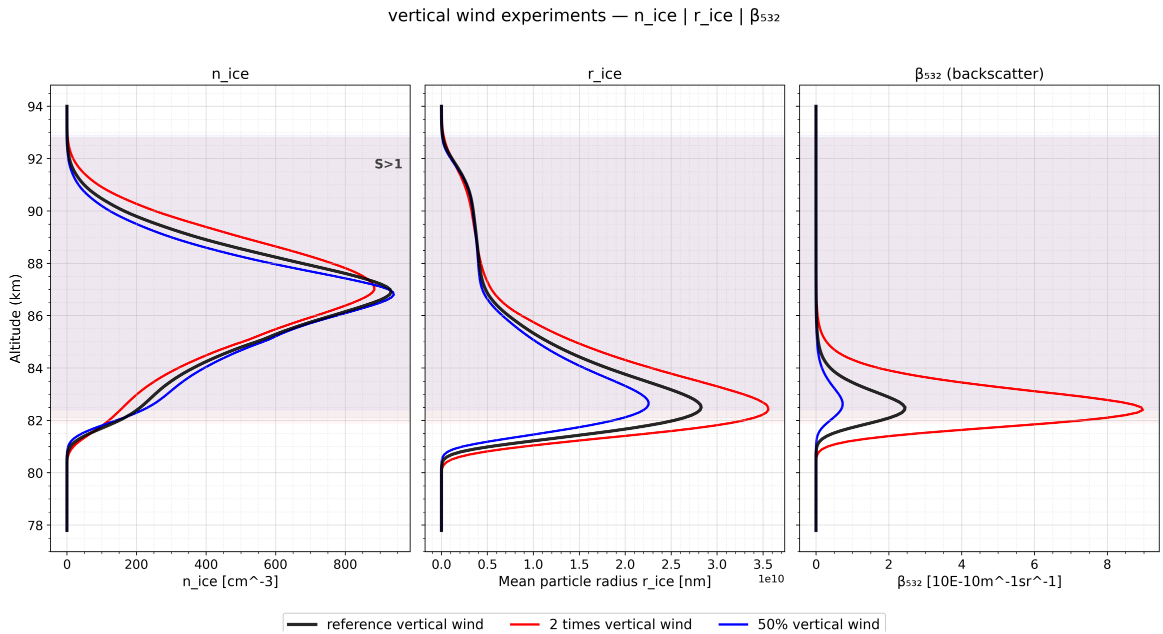


Figure 3.10: Vertical profiles of ice particle number density (n_{ice}), mean particle radius (r_{ice}), and optical backscatter coefficient (β_{532}) for three vertical wind conditions.

the H_2O profile but reshapes the entire background structure in which NLCs form.

3.2.2. Impact on NLC properties

The profiles of ice particle number density (Figure 3.10) show a clear sensitivity to changes in vertical wind. In contrast to turbulence, which broadens the vertical distribution and noticeably alters the peak value of n_{ice}, increasing the vertical wind mainly shifts the entire profile upward without significantly changing its shape. As the vertical wind lifts both the ice formation zone and the particles themselves, the altitude of maximum number density also rises. Ice formation beginning at higher altitudes implies two things: (1) more water vapour is available at those heights, and (2) once formed, ice particles remain longer within the now-broadened ice-formation region. Both effects support stronger growth. This upward shift is also evident in Figure 3.11, where the top row shows that the ice-forming layer becomes thicker and extends to higher altitudes (up to 88–90 km) under stronger vertical wind. These findings support earlier interpretations that NLC formation is controlled mainly by local supersaturation and available condensation nuclei, while vertical winds modulate the vertical placement and residence time of the particles (Rapp & Thomas, 2006; Berger & Zahn, 2007).

As discussed in the turbulence case, controlling the sedimentation speed directly influences particle growth by modifying the time an ice particle spends inside the ice formation zone. Unlike turbulence, which randomly mixes particles across the S=1 boundary, increasing vertical wind consistently increases particle size across all altitudes. The most notable differences occur near the altitude where the mean radius peaks. With reduced vertical wind (0.5× reference), the mean radius stays below 25 nm, while under enhanced vertical wind (2× reference), values exceed 35 nm. Two mechanisms explain this: (1) stronger vertical winds transport more water vapour upward, enhancing supersaturation and lowering the

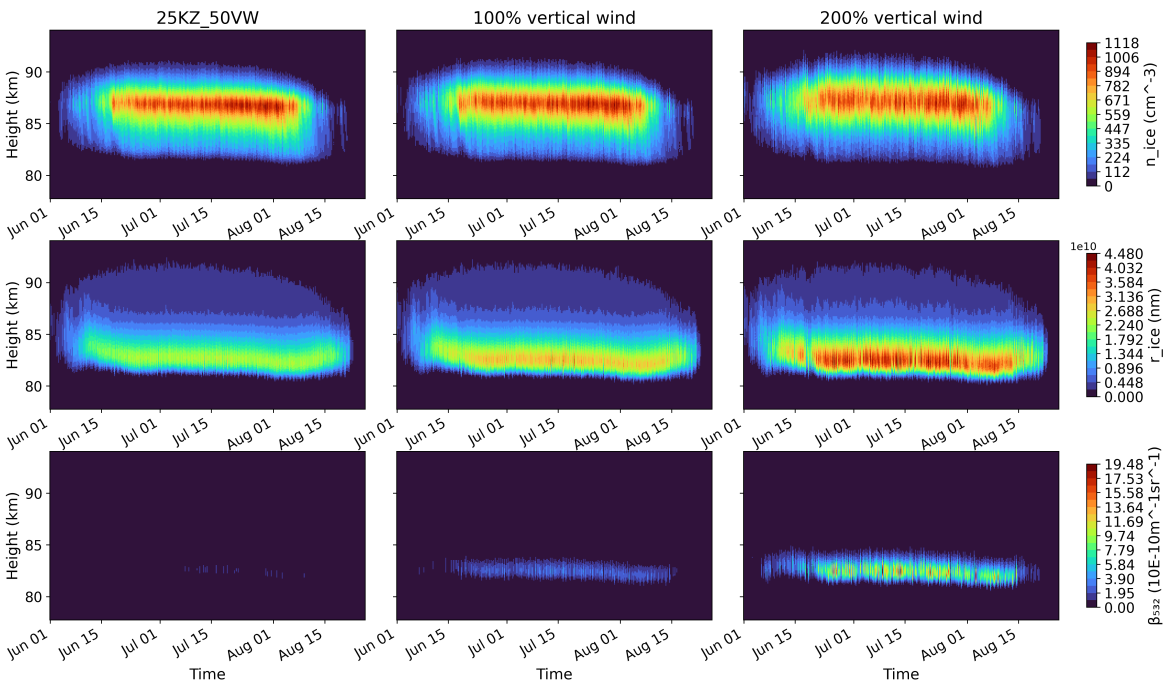


Figure 3.11: Spatiotemporal variation of key NLC properties under different vertical wind conditions. Columns represent vertical wind values of 50%, 100%, and 200% relative to the reference case. Rows show, from top to bottom, the number density of ice particles (n_{ice} [cm^{-3}]), mean particle radius (r_{ice} [nm]), and optical backscatter coefficient (β_{532} [$10\mathrm{E}^{-10}$ $\mathrm{m}^{-1}\mathrm{sr}^{-1}$]).

S=1 boundary, and (2) the balance between upward wind and downward sedimentation shifts. For example, vertical wind speeds in MIMAS near 83 km are about 2 cm/s (Figure 3.7), while typical sedimentation velocities of larger (50 nm) ice particles are around 10 cm/s (Kiliani et al., 2013). Doubling the upward wind significantly reduces the net downward speed, extending the time available for growth in the maximum growth region (83–84 km). This interplay between vertical transport and microphysical growth strongly influences the overall ice water content (Rapp & Thomas, 2006).

As more ice particles appear at higher altitudes when vertical wind is increased, the lower sublimation zone and the altitude of maximum radius shift slightly downward (about 200 m). The broadened formation region and longer residence time produce noticeably larger particles. These effects are even clearer when examining the brightness curves in Figure 3.10 (right panel). Under weak vertical wind (0.5×), NLCs become almost invisible, with minimal backscatter. In contrast, doubling the vertical wind increases the average optical brightness by more than an order of magnitude (see Figure 3.11). Much larger particles are now present, with many exceeding the Rayleigh–Mie transition threshold.

The NLC brightness profiles demonstrate a strong sensitivity to vertical wind variations. Brightness increases substantially with increasing vertical wind, and the vertical extent of the bright layer also grows. This enhancement is primarily caused by changes in particle radius. Since the backscatter coefficient β is approximately proportional to the 5^{th}–6^{th} power of particle radius, even a modest reduction in radius leads to a dramatic drop in brightness. Doubling the vertical wind produces roughly a 26% increase in peak particle radius, resulting in nearly a 260% increase in maximum brightness. These results highlight how mesospheric dynamics regulate the optical visibility of NLCs: periods of strong upwelling, often linked to tides and gravity waves can create brighter, more persistent, and optically dense cloud layers (DeLand et al., 2003; Kiliani et al., 2013).

4
Conclusion

The aim of this thesis is to understand how mesospheric dynamical forcing, especially small-scale turbulence and large scale vertical wind controls the formation and evolution of noctilucent clouds (NLCs) in a realistic summer polar environment. By coupling LIMA background fields with the Lagrangian microphysical scheme of MIMAS, the simulations resolve the growth, motion, and freeze drying signatures of individual ice particles. The results show that even moderate changes in K_{zz} or vertical wind substantially modify particle sizes, number densities and the optical brightness of NLCs. Here, these findings are interpreted in the broader context of observational evidence, laboratory measurements, and previous modelling studies.

Turbulence mainly influences the fine structure of the cloud. When K_{zz} is increased, particles spend less time in the narrow altitude region where most growth occurs. As a result, the cloud becomes thinner, the mean particle radius decreases, and the overall brightness weakens. This behaviour is consistent with lidar and satellite observations, which show that enhanced mixing tends to reduce cloud contrast and shift particle sizes toward smaller values (Pérot et al., 2010; Langowski et al., 2018; Feofilov & Petelina, 2010; Zhang et al., 2022). These similarities indicate that the model captures the essential physical link between turbulence, reduced growth time, and diminished optical strength.

Vertical wind acts in a different and more systematic way. A stronger updraft brings additional water vapour to the coldest region, counteracts sedimentation, and keeps particles inside supersaturated layers for longer. In the $2 \times w$ experiment, mean radii increased by roughly twenty percent and NLC brightness strengthened by more than a factor of three, reflecting the strong radius dependence of light scattering. This upward shift of the vapour–particle system mirrors semidiurnal tidal signatures observed in lidar data (Baumgarten et al., 2010; Kiliani et al., 2013), and the enhanced freeze drying below the cloud agrees with satellite observed hydration layers beneath active NLCs (Feofilov & Petelina, 2010). Together, these comparisons confirm that vertical motion is a primary control on cloud altitude, brightness, and particle longevity.

The model behaviour aligns well with laboratory studies on deposition growth, which show that nanometre scale particles grow only slowly at mesospheric temperatures and require sustained supersaturation to reach detectable sizes (Duft et al., 2019). As the increasing turbulence reduces the time particles spend near the $S = 1$ boundary where more than 80

Homogeneous nucleation is theoretically possible only at extremely high supersaturations (Murray & Jensen, 2010; Tanaka et al., 2022), far beyond typical polar mesopause values.

Its exclusion from this study is therefore consistent with both observations and previous MIMAS-based results. Heterogeneous nucleation on meteoric smoke remains the dominant pathway for NLC formation under realistic conditions.

Comparisons with other modelling systems (Wilms et al., 2016) show that the magnitude and nature of the NLC response found here lie well within expectations from independent frameworks. The weakening of freeze drying under strong mixing also agrees with the steep vertical structure of K_{zz} inferred from rocket observations (Lübken et al., 2018). These consistencies demonstrate that the sensitivities identified in this thesis are robust features of the coupled mesospheric dynamics, microphysics system.

Overall, the results show that turbulence and vertical wind shape NLCs through two distinct physical pathways: turbulence smooths vertical gradients and shortens effective growth times, while vertical wind coherently shifts particles and vapour through the cold region, enhancing growth and brightness. Their combined influence explains many observed variations in NLC visibility that occur even without changes in temperature or background water vapour.

In conclusion, mesospheric dynamics do not merely "modulate" NLCs they fundamentally constrain how ice particles grow, sediment, and sublimate. By linking model results with observational records and laboratory evidence, this thesis provides a physically grounded explanation of NLC sensitivity to dynamical forcing and highlights the importance of representing realistic mesospheric variability in future climate simulations.

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